

④ Database

A database is a collection of organized data stored in a structured way so that it can be easily accessed, managed and updated.

Database is many type

① Relational Database (RDBMS)

Data is stored in tables and tables are related to each other.

Ex. MySQL, Oracle

② NoSQL Database

Data is stored in non-table format like JSON, Key-Value etc.

Ex. MongoDB

③ Hierarchical Database

Data is stored in tree structure (Parent $\rightarrow$ child)

① Network Database

Data is connected like a network
(Many-to-many relation)

② What is DBMS ⇒ Database Management System is Software used to create, manage and control database.
Ex. MySQL, Oracle

③ What are the features of database?

- Data is organized ✓
- Easy to access data ✓
- Data Security ✓
- Data Consistency ✓
- Avoid data duplication
- Backup and recovery

④ What is the difference btw DBMS/ RDBMS

DBMS ⇒ DBMS stores data as a file
→ supports Single user only
→ Data fetching is slower for the complex and large amount of data.
Ex. XML

RDBMS ⇒ RDBMS ~~stores~~ Data is stored in the form of table
⇒ It support Multiple users
→ Multiple data elements can be accessed at the same time
→ Data fetching is rapid because of its relational approach
Ex. MySQL, SQL

Components of database

Table ⇒ Stores data

Row ⇒ one record / tuple

Column ⇒ one field / Attribute

redundancy $\Rightarrow$ data redundancy means duplicate data stored in database

data integrity $\Rightarrow$ means data is correct and consistent.

Schema $\Rightarrow$ Schema is the structure of database.

Metadata $\Rightarrow$ Metadata is data about data (Information about structure).

Entity $\Rightarrow$ An entity is a real-world object that we store data about.

Note $\Rightarrow$ Entity is an object (like student, and employee).

Note

Attribute is Column, Tuple is row and

Entity is real world object represented in a table.

ER MODEL:-

ER Model is used to design a database using entities, attributes and relationships. using ER diagram

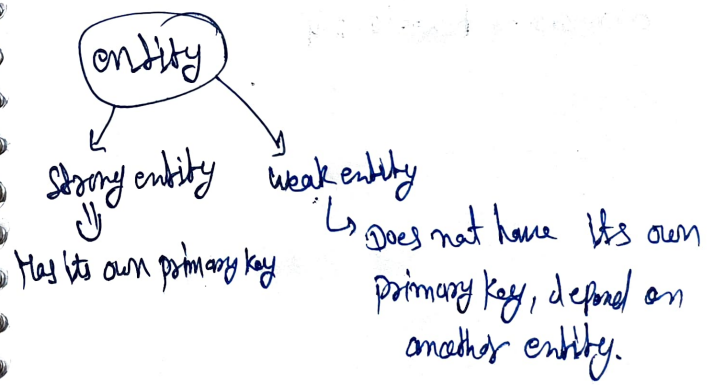
$\Rightarrow$ ER Model helps us to design database structure.

Components of ER Model:-

① Entity $\Rightarrow$ is real world objects about which we store data Ex. student, employee

② Attribute $\Rightarrow$ An attribute is a property of an entity Ex. name, age

③ Relationship $\Rightarrow$ Shows how two entities are connected.
Ex. student $\rightarrow$ enroll $\rightarrow$ course



type of Relationship

- ① one to one
- ② one to many
- ③ many to many

Q. What is ER diagram?

ER diagram is a visual representation of database using entities, attributes and relationships.

Q. Symbols used in ER diagram?

Rectangle → Entity
oval → Attribute
Diamond → Relationship

SAL

SAL (Structured Query language) is used to communicate with database.

⇒ SAL is used to store, retrieve, update and delete data in a database.

SAL command types

① DDL (Data Definition language) is used to define and manage the structure of database.

Ex. ① Create

② ALTER → used to modify an existing table.
deleted table structure

③ DROP → used to delete a table or database completely.

delete only data safe structure. ④ TRUNCATE → used to delete all data from a table but keep the structure.

⑤ RENAME → used to change the name of a table.

Rename table oldName to newName.

⑥ auto-commit

↳ in SQL, commands are auto commit, they cannot be rollback.

② DML (Data Manipulation language) is used to Manage data in a table.

① INSERT → is used to add new data into a table

Insert into table values ()

② Update → is used to Modify existing data in a table

update table set columnname =
where condition

③ Delete → is used to remove data from a table

Delete → remove specific rows (with where)

Truncate → remove all rows (no where but keep the structure)

Note

① Delete without where → It delete all record from the table as like truncate

② Can we rollback DML?

Yes, DML Command can be rolled back (using Rollback)

③ DCL (Data Control language) → is used to Control access to database.

⇒ DCL is used to give and remove permissions.

① Grant → is used to give permission to a user.
↳ Grant allows a user to perform action on database

Grant privileges ON table to user (~~table~~)

② Revoke → is used to remove privileges from a user.

Revoke privileges ON table from user

③ Role → A Role is a Collection of privileges.

Create Role RoleName;

Grant privileges ON table to RoleName;

Grant RoleName To user.

Use DCL:

↳ used for Security and access Control.

↳ used for Controlling user permission in database

① TCL (Transaction Control language) is used to Manage transaction in a database.

→ TCL Control changes Made by DML Command.

Transaction → A transaction is a group of SQL operation executed as a single Unit.

① Commit → is used to Save change permanently
↳ All change are saved in database.

② RollBack → is used to Undo change. But RollBack can be used before commit.
↳ Cancels the Change and return the database to the previous state.

③ Savepoint → is used to create a point to roll back.

→ Rollback To SPI
Use TCL → TCL is used to Maintain data Consistency and Control transactions.

⇒ Start transaction → It begins a transaction.

⇒ @ auto Commit → AutoCommit is a Mode where every SQL statement is automatically Saved.

⇒ Set AutoCommit=0

↳ is used to disables autoCommit.

⇒ Set AutoCommit=1

↳ is used to enables autoCommit.

Q Why do we disable Auto Commit?

↳ To Control transaction manually using Commit and Rollback.

Q What happens if AutoCommit is ON?

⇒ every change is saved immediately, and we cannot rollback.

ACID property → ensure reliable and safe database transaction.

A → Atomicity → Transaction is all or nothing

C → Consistency → Data Valid

I → Isolation → Transaction Independent

D → Durability → ~~data~~ once committed, data is permanent.

⑤ DQL (Data Query Language) is used to retrieve data from a database.
⇒ DQL is used to fetch data.

select ⇒ is used to get data from a table.

where ⇒ is used to filter data

orderby ⇒ is used to sort data

distinct ⇒ is used to remove duplicate value.

Limit ⇒ is used to restrict number of rows.

aggregate function ⇒ used to perform calculation on data

Group by ⇒ used to group rows with same value

Having ⇒ is used to filter grouped data
↳ filter group

Execution order ⇒

select → where → orderby → Groupby → Having.

⇒ select में किया गया change, उस data में बदलाव नहीं करता है।

NOTE

Limit ⇒ is used to restrict the number of rows returned

Limit 5;

↳ Show only 5 rows

Limit 1, 5;

skip ↳ Next 5 rows return

↳ skip rows are called offset

↳ Limit 5 offset 2

↳ skip 2 rows then return 5 rows.

Count(*) ⇒ Count all rows (Including Null)

Count(Column) ⇒ Count only non-null value in that Column.

Q Can we use limit with orderby;

Yes, Limit is often used with orderby to get top records

Data Model

Data Model :- a data Model is a way to define how data is stored, organized and related in a database.

⇒ Data Model tells us structure of database.
there are three level

① Conceptual Data Model ⇒ Conceptual Model shows what data is needed in the database

⇒ It is high-level view of database

Key ⇒ what data we want
→ No technical details

② Logical Data Model ⇒ Logical Model show how data is structure logically.

⇒ It defines tables, columns, and relationship.

Key : Table, Attributes, keys

③ Physical Data Model ⇒ physical Model show how data is stored in database.

⇒ It is actual implementation in database

Key ⇒ Storage, Datatypes, Indexes, Performance

Note :-

A data Model defines how data is stored and organized in a database. It has three levels :-
Conceptual (what data is needed)
Logical (how data is structured)
Physical (how data is stored)

Keys

⇒ Key are attributes used to Identify records and create relationship in a table

Types of Keys

① primary Key ⇒ primary Key uniquely ~~Identify~~ identifies each record in a table.

⇒ Must be Unique

⇒ Can not be Null

② foreign Key ⇒ It create relationship b/w tables.

③ Candidate Key $\Rightarrow$ Candidate keys are all possible keys that can uniquely identify a record.

$\Rightarrow$ Multiple Unique Key available

④ Super Key $\Rightarrow$ Super key is a set of attributes that can uniquely identify a record.

$\Rightarrow$ It may contain extra attributes.

⑤ Composite Key $\Rightarrow$ Composite key is made by combining two or more columns.

operators

operators are symbols used to perform operation on data.

$\rightarrow$ operators help to compare, calculate or filter data.

Types of SQL operators

① Arithmetic operators $(-, +, \times, \div, /)$ used for Mathematical calculation

② Comparison operators $\Rightarrow$ used to compare value
 $\hookrightarrow (=, !=, <, >, <=, >=)$

③ Logical operators $\Rightarrow$ used to combine multiple conditions.

(AND, OR, NOT) if used (AND, OR) $\frac{1}{2}$ executes

④ Special operators

① IN $\rightarrow$ is used to match multiple values.

② BETWEEN $\rightarrow$ is used to select values within a range.

$\Rightarrow$ Between with not use 'or'

③ LIKE $\Rightarrow$ is used to search for a pattern in data.

MySQL

MySQL is a open-source relational database Management System used to store and Manage data

⇒ It is Software

SQL ⇒ Language to query database

MySQL ⇒ Software that use SQL

String Function:-

① Upper() → Convert text into uppercase

② Lower() → Convert text into lowercase

③ Length() → Return number of characters in string
↳ Bytes count. → Str

④ char_length() → Return number of characters in string
↳ include space

⑤ Concat() ⇒ Join two or more strings

⑥ Concat_ws() ⇒ Join string with a separator
Concat_ws('-', '2026', '04', '03') ⇒ 2026-04-03

⑦ Substring() ⇒ Extracts part of a string
Substring('Krishna', 1, 4) ⇒ 'Kris'
↳ Start and

⑧ Trim() ⇒ Remove space from start and end of string

⑨ replace() ⇒ is used to replace part of a string with another string

Replace('Krishna', 'Krishna', 'Yash')
↳ इसकी जगह Yash होगा।

Replace('I Love Java', 'Java', 'SQL')
↳ इसकी जगह

⑩ Padding $\Rightarrow$ is used to add extra characters to a string
Two type $\rightarrow$ LPAD('SQL', 6, 'x') $\Rightarrow$ xxxSQL
 $\rightarrow$ RPAD

⑪ Distinct $\Rightarrow$ is used to remove duplicate values.

⑫ Group_Concat() $\Rightarrow$ is used to combine multiple row value into one string.

⑬ COALESCE() $\Rightarrow$ return the first non-null value from a list of value.

COALESCE(email, 'Noemail')
 $\rightarrow$ if email null then Noemail show

⑭ IfNull $\rightarrow$ is used to replace null with a specified value.

$\Rightarrow$ IfNull(email, 'NOTE')

Nullif $\Rightarrow$ if two value are equal then return otherwise first value.

Date Functions

① Now() $\rightarrow$ current date and time

② curdate() $\rightarrow$ current date

③ curtime() $\rightarrow$ current time

④ Date(now()) $\rightarrow$ only date for timeline

⑤ YearMonth() $\rightarrow$ only month for

⑥ datediff() $\rightarrow$ Return difference b/w two date (in day)

⑦ Date_Add() $\rightarrow$ Add day/month/year to a date

Date_Add('2026-04-09', Interval 5 Day);

⑧ Date_Sub() $\rightarrow$ Subtracts days/month/year to a date.

⑨ Year(), Month(), day() $\rightarrow$ extract Year, month, day.

Maths

① ABS() → absolute (positive) value of a Number

② Round() → Round a number to nearest value

Round(12.56) → 13

Round(12.34, 1) → 12.3

③ Ceil() → Return smallest integer greater than or equal number

Ceil(12.3) → 13

Floor → largest integer less than or equal to number

Floor(12.3) → 12

④ MOD → remainder of division

⑤ Power → Return Power of Number

Power(2, 3) → 8

⑥ Sqrt() → Return Square root of a number

Sqrt(16) → 4

⑦ Truncate() → cut decimal number to a fix number of digit

Truncate(12.3456, 2) → 12.34

function:- block of code → code reusable

→ pre defined and user defined

→ pre defined → Scalar function

↳ applied for each row and result will be given for each row

→ String, number, date related function.

Groupby:-

Groupby groups of similar rows and is used with aggregate function to get Summary results.

⇒ जिस Column को Groupby करने को उसे Select में लिखें।

⇒ Groupby → Group data
↳ Summary result

⇒ Distinct → Remove duplicate
↳ Unique rows.

Having ⇒ Having is used to filter grouped data.
↳ and is used after group by

Normalization

⇒ Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity.

- remove duplicate data
- avoid data inconsistency
- improve database efficiency

Normal form ⇒ Normal forms are steps of normalization.

1NF ⇒ Table should have atomic value
↳ no multiple value in a single column.

2NF ⇒ Table should be in 1NF.
↳ data should be depend on full primary key.

It use → Remove partial dependency

3NF ⇒ Table should be 2NF and no transitive dependency should exist.

means → Non-key attribute should not depend on another non-key attribute.

BCNF (Boyce-Codd Normal form) :-

BCNF is a advance form 3NF where every determinant must be a candidate key.
↳ it is d

use → remove anomalies that are not handled by 3NF.

Join

Join is used to combine data from two or more tables based on a related Column.

Join are many type

① Inner Join $\Rightarrow$ Return only Matching records from both tables.

② Left Join $\Rightarrow$ Return all record from left table and Matching from right.

③ Right Join $\Rightarrow$ Return all record from Right table and Matching from Left.

④ Full Join $\Rightarrow$ Return all record from both side
 $\hookrightarrow$ full join not support MySQL

⑤ Cross Join $\Rightarrow$ return all possible combinations of row from two table
table 1 (row 2)
table 2 (row 5) Result = $2 \times 5 = 10$

⑥ Natural Join $\Rightarrow$ automatically join table based on same Column name.

Join $\Rightarrow$ join work combin table, and faster with easy Readability.

Subquery $\Rightarrow$ Subquery work Nested query. It sometimes slower and complex Readability.

Subquery

A subquery is a query inside another query

use: To get data based on result of another query

Types of Subquery

① Single-row Subquery $\Rightarrow$ Return only one value

② Multi-row Subquery $\Rightarrow$ Return Multiple values

③ Correlated Subquery $\Rightarrow$ Run once for each row of outer query

$\Rightarrow$ Correlated Subquery $\Rightarrow$ Select, where, from

all, any, IN, Exist

IN $\Rightarrow$ IN is used to ~~Match~~ Match Value from a list or Subquery.

As like (or operator)

ANY $\Rightarrow$ ANY Compares a Value with any value returned by Subquery.

IN = ANY

ALL $\Rightarrow$ ALL Compares a Value with all Value of Subquery.

(all Condition true)

Exist $\Rightarrow$ Exist checks whether Subquery return any row.

Window Function

A window function perform calculation across a Set of rows without grouping them.

Group by $\Rightarrow$ rows Combine
Window $\Rightarrow$ rows Same (no merges)

Syntax

Select Column,

function() over (partition by column order by
Column) from table;

Window function type

① Ranking function

① Rank() $\rightarrow$ Same Rank for Same Value
 $\hookrightarrow$ But skips number after tie.

② Dense-Rank() $\rightarrow$ Same Rank for Same Value
 $\hookrightarrow$ But it not skips number after tie.
 $\hookrightarrow$ No gap

① Row-Number()

- Given Unique Number for each row
- No duplicate
- No skip

② Aggregate Functions

③ Value Functions:- Previous/Next Value

① LAG() → Give previous row Value

② LEAD() → Give Next row Value

→ ③ NTILE(^{group need}) → divides rows into equal group (buckets)

use → Top Performers
→ Segmentation

over() → define the window for a function
partition by → divides the data into groups without combining row.

⇒ Row between unbounded preceding and unbounded following
unbounded preceding → first row
unbounded following → Last row
→ Full table

Running Sum → unbounded preceding + current row
Future Sum → current row + unbounded following

1. preceding + current row → current + previous row
current row + 1 following → current + Next row
1 preceding + 1 following → previous + current + next.

Constraint

Constraints are rules applied on table column to ensure data accuracy and integrity.

Type of Constraints

- ① NOT NULL → Column cannot be empty
- ② UNIQUE → Ensure all Value are different
But allow Multiple Null and NOT work with Default.
- ④ primary Key → Uniquely Identifies each record in a table.
↳ Unique + Not Null

③ foreign key:- Link one table to another table

① check $\Rightarrow$ Ensures condition is satisfied.

② Default $\Rightarrow$ Assigns default value if no value is given.

CTE (Common Table Expression)

$\hookrightarrow$ A CTE is a Temporary result set used within a query.

Syntax

with cte_name as (query)

Select * from cte_name;

Use:- ~~Complex query to simple~~

- | | |
|-----------------------|---------------------------------|
| ① Make query simple | $\rightarrow$ Salary analysis |
| ② Improve Readability | $\rightarrow$ Department report |
| ③ Reuse Data | $\rightarrow$ Rankings |
| ④ Replace Subquery | $\rightarrow$ Hierarchy data |
| ⑤ Recursive problems | |

Recursive cte:-

Recursive cte is a cte that calls itself.

use $\rightarrow$ number series
tree/hierarchy
parent-child data

Structure

$\rightarrow$ Base Case $\rightarrow$ Starting Value
 $\rightarrow$ Recursive Case $\rightarrow$ repeat Logic

with recursive name as (query
condition New)

Union all

query + join recursive

Index

$\rightarrow$ An index is used to speed up data retrieval.
 $\rightarrow$ fast searching tool

Use

$\rightarrow$ query fast
 $\rightarrow$ Improve search performance
 $\rightarrow$ Reduce table scan

Syntax

Create Index Index-name
ON table (Column-name);

Type of Index

- ① Single Column Index
- ② Composite Index (Multiple Column par)
- ③ Unique Index → Not allow duplicate
- ④ Clustered Index
 - ↳ Data stored physically in stored order
 - ↳ one table
 - ↳ UK → primary key.
- ⑤ Non-clustered Index
 - ↳ Separate structure
 - ↳ Multiple table
 - ↳ Search column not primary/key.

When to use Index

- Large table
- where condition
- Join, Order by frequently.

View

~~Def~~ → A View is a virtual table based on query
→ Table → real data
View → us data ka shortcut/snapshot

Syntax

Create View Viewname as query;

Note → It ~~do~~ does not data stored physically.
It show data form one or more table

View Update Condition:

- Single table
- No Groupby, No Join, No aggregate

① Simple View :- View create from one table without complex operations.

Ex. No join, no function

② Complex View → View created using multiple table or functions.
Ex → join, Groupby function

② Materialized View :-

A View that stores data physically.

With Check option

↳ Ensure data inserted/update through View satisfies View Condition.

Trigger :- A trigger is a set of SQL Statement that automatically runs when an event occurs.

- Insert
- Update
- Delete

Stored procedure :- A stored procedure is a saved SQL code that can be executed anytime.

Note :- A trigger is automatically executed when an event occurs, whereas a stored procedure is a saved SQL program that must be executed manually.

PL/SQL

PL/SQL ⇒ PL/SQL is a procedural extension of SQL
SQL + programming

Syntax

```
Declare
  -- Variables
Begin
  -- Logic + SQL
END;
```

point ⇒ DBMS_output.put_line ();

Use :- Stored procedure

- Trigger
- Complex logic
- Automation
- It is a portable

Note PL/SQL is a procedural language that extends SQL by adding programming features like variables, loops, and conditions.

PL/SQL have ~~generally~~ generally three block.

① Declaration block

→ Keyword → declare
→ Variable, Tiger, cursor, function

② Executable block ⇒ → ~~the~~

→ Begin
→ E → logical
→ end

③ Exception block ⇒ error Handl.

Record ⇒ A record is a collection of related fields stored as one unit.

Ex: ① Table ke same structure ka record

Declare

emp_rec employees % Rowtype;

Begin

emp_rec.name := 'Ravi';

end;

② User Define.

~~SQL~~ Declare

Type emp_type is Record (name varchar(20),
Salary Number);

emp_rec emp_type;

Begin

emp_rec.name :=

emp_rec.Salary

end;

Cursor

A cursor is used to fetch data row by row from a query.

Type

① Implicit cursor

↳ Automatically create

② Explicit cursor

↳ Manually create

Cursor Attribute

% found ⇒ row found

% Not found ⇒ row Not found

% Rowcount ⇒ Count Row

% Isopen ⇒ open → Yes/No.

Step

① Declare

② open

③ fetch

④ close

Use:- → Row by Row Processing

→ Complex logic

→ Reports

→ Data transformation.